Monitoring climate change

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Abstract. The students studied a scenario about global warming and climate change. The purpose of the scenario was to encourage students to discover the factors that lead to global warming and climate change. The didactic approaches followed by the scenario were: the exploratory, the experimental, the collaborative and the problem solving method. The scenario followed the steps: care, learn and act. The scenario made students aware of a real problem with local, global, present and future implications

The students studied the scenario about global warming and climate change. The purpose of the scenario was to encourage students to discover the factors that lead to global warming and climate change. It is accepted that the burning of fossil fuels leads to global warming which contributes to climate change. The students worked on a real problem and tried to find solutions. The aim of the scenario was to make them environmentally oriented citizens. Climate change refers to the increase in the earth's average temperature and the rain distribution over the last 50 years. This increase is linked to the greenhouse effect and the gases present in the atmosphere such as carbon dioxide, water vapor and methane. Human activities contribute to the greenhouse effect. Climate change is nevertheless a millennia on going process with man nowdays contributing to it. Increasing earth population, deforestation, change of land use, inevitably lead to climate change. The purpose of the scenario was for the students to understand that their actions and daily habits contribute to global warming and reinforce climate change. The didactic approaches followed by the scenario were: the exploratory, the experimental, the collaborative and the problem solving method. The scenario followed the steps: care, learn and act. In the "care" stage students' perceptions of climate change were explored. In the "learn" stage they watched appropriate videos and studied articles about warming, weather, climate and the physical processes involved. They watched some videos together with their parents, answered quiz questions for better understanding the problem and also discussed and exchanged opinions on the specific topic. The students met with two specialist scientists who presented them the scientific problem and they discussed with them the prospects of the future. In the "act" stage they produced material that presents the problem and proposed solutions in order to limit it. They constructed an electronic weather station which measures temperature, relative humidity, surface pressure and rain height and publishes the data on the internet. Additionally, they developed a questionnaire to test their classmates' knowledge and perceptions on environmental and energy consumption issues. Furthermore, they prepared an article for the local newspapers and a proposal to the municipality of the city proposing how to better transfer conditions with public transport as well as proposals to improve the city's green spaces. An important aspect of the scenario was the comprehension and use of the scientific method as an approach to solve environmental problems. The scenario made students aware of a real problem with local, global, present and future implications.

References

1. https://www.connect-science.net/about-the-project/
2. https://www.connectscience.org/

3. https://www.connect-science.net/scientific/global-warming-and-chemical-pollution/
4. https://www.livescience.com/43296-what-is-stem-education.html
5. https://lastminuteengineers.com/bme280-esp32-weather-station/?utm_content=cmp-true
6. https://pantou.sites.sch.gr/ClimateChange/testthis.htm Monitoring climate change in greek

Youtube : Can you fix climate change? (https://www.youtube.com/watch?v=yiw6_JakZFc)

Website : COP26 (https://ukcop26.org)

Website : There is no climate emergency https://clintel.org/

Educational Materials to download or watch

To watch a video on you tube with Greek subtitles we must first select subtitles and then from the automatic translation option Greek

The following link contains material that can be used by the educator to understand the natural phenomena and human interventions that can cause or contribute to climate change.

https://drive.google.com/drive/folders/1LJUkgrEsXozGYidc7sLo2w6VPuERsi7P?usp=sharing

Video

1A . Very accurate recording of CO 2 emissions in Heraklion

https://www.youtube.com/watch?v=Y_nK4Afmon0

1B. Foskolos: "climate change exists, it is not due to carbon dioxide"

https://www.youtube.com/watch?v=IvuCEROEjI8 **abnormal phenomenon**

1C. The anomalous greenhouse effect

https://ec.europa.eu/environment/archives/youth/air/air_abnormalgh_el.html

1D. Climate Change In Simple Words - Konstantinos Kartalis

https://www.youtube.com/watch?v=p7YldLvolOw

1E . _ Emission reductions from the pandemic have had unexpected effects on the atmosphere

https :// www . jpl . nasa . gov / news / emission - reductions - from - pandemic - had - unexpected - effects - on - atmosphere (use google translate)

1F Global warming and climate change: Myth or reality?

1Z S. Kamenopoulos : The exaggerations about climate change and hydrocarbons

1.There is 'nothing bad' about increasing carbon dioxide: William Happer

https://www.youtube.com/watch?v=Wbs8whRlzn0

2.The Maths of Climate Change

https://www.youtube.com/watch?v=w4O4jK-lZrI&t=329s

3.Andy May: “CO2-driven climate models of the IPCC are inadequate”

https://www.youtube.com/watch?v=6aNkmXArlZk

4. Matt Ridley on How Fossil Fuels are Greening the Planet

https://www.youtube.com/watch?v=S-nsU_DaIZE

5. Irrefutable points on Global Warming, apparently

https://www.youtube.com/watch?v=JHN7PrU5ZYw

Links translated

1.There is "nothing wrong" with increasing carbon dioxide: William Happer

https://www.youtube.com/watch?v=Wbs8whRlzn0

2. The mathematics of climate change

https://www.youtube.com/watch?v=w4O4jK-lZrI&t=329s

3. Andy May : May : " IPCC climate models based on CO 2 are inadequate"

https://www.youtube.com/watch?v=6aNkmXArlZk

4. Matt Ridley on how fossil fuels are greening the planet

https://www.youtube.com/watch?v=S-nsU_DaIZE

5. Arguable points about global warming, obviously

https://www.youtube.com/watch?v=JHN7PrU5ZYw